Lecture 1: Machine Learning in 2026 and a Mathematical Review + Bridge

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WESTLAKE UNIVERSITY | SCHOOL OF ENGINEERING

What you should be able to explain after today

From examples to a learning problem

1. Identify inputs, targets, parameters, loss, training, and evaluation.
2. Explain how representations turn numbers into task-relevant features.

From modern systems to mathematical review

3. Relate algorithms, data, compute, and systems without assuming unlimited scaling.
4. Review geometry and uncertainty, then bridge gradients and composition to later lectures.

45 min Modern ML
data $\rightarrow$ models $\rightarrow$ decisions

$\rightarrow$

45 min Review + bridge
linear algebra review;
two calculus recaps

$\rightarrow$

45 min Probability review
distributions, conditioning, Bayes

Check yourself

Check yourself (rapid check)

Training loss falls while test error rises. Which part of the pipeline is warning us?

Commit to an answer before advancing the slide.

A learning system connects data to decisions

training examples $\mathcal{D} = \{(\mathbf{x}_n, y_n)\}_{n=1}^{N}, \qquad \mathbf{x}_n \in \mathcal{X} \subseteq \mathbb{R}^D,\ y_n \in \mathcal{Y}$

INFERENCE

input $\mathbf{x}$ $\rightarrow$ model $f_{\boldsymbol{\theta}}$ $\rightarrow$ prediction $\widehat{y}$ $\rightarrow$ decision / action

TRAINING

prediction $\widehat{y}_n$ + target y_n $\rightarrow$ loss $\ell(\widehat{y}_n, y_n)$ $\rightarrow$ parameter update $\boldsymbol{\theta} \leftarrow \mathrm{opt}(\boldsymbol{\theta}, \nabla \ell)$

Evaluation is separate: test the trained inference path on held-out data, relevant shifts, and deployment costs.

Check yourself (rapid check) — reveal

Question: Training loss falls while test error rises. Which part of the pipeline is warning us?
Answer: Evaluation exposes a growing generalization gap; lower training loss alone is not the goal.

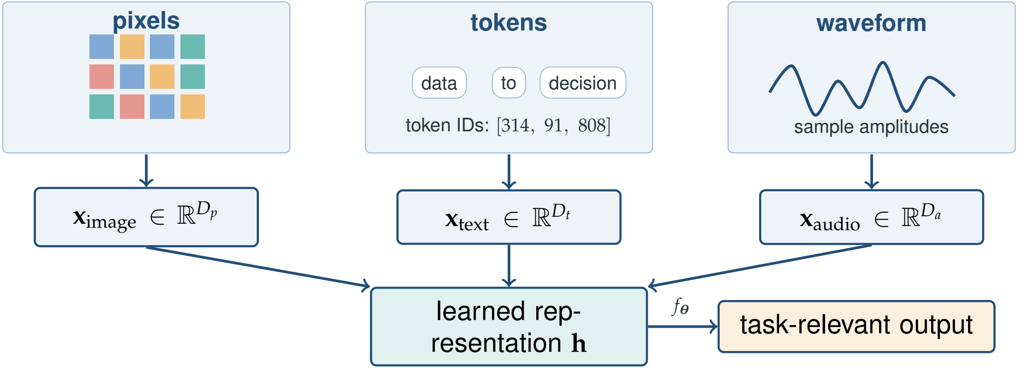
Rapid check

Rapid check

If two inputs are close as raw vectors, must they mean the same thing?

Commit to an answer before advancing the slide.

Inputs are numbers before they become meaning



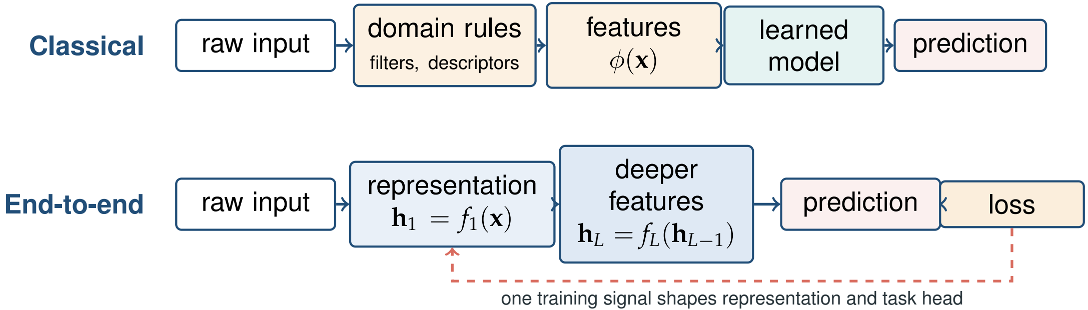
Rapid check — reveal

Question: If two inputs are close as raw vectors, must they mean the same thing?
Answer: No. Coordinates carry measurements; raw distance depends on the representation and may not align with semantics.

Representation learning reduces hand-designed interfaces

Classical raw input → domain rules (filters, descriptors) → features $\phi(\mathbf{x})$ → learned model → prediction

Representation learning reduces hand-designed interfaces



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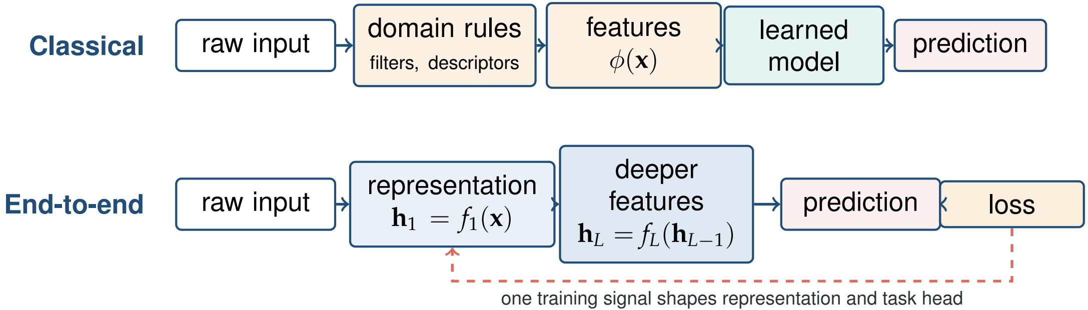


The interface moves; it does not disappear

Learned features reduce manual feature engineering, while people still choose data, architecture, objectives, constraints, and evaluation.

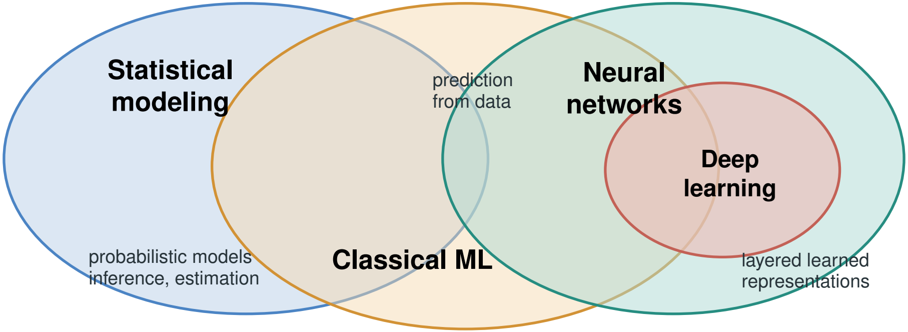
Optional rapid check

Optional rapid check

Is deep learning outside statistical modeling?

Commit to an answer before advancing the slide.

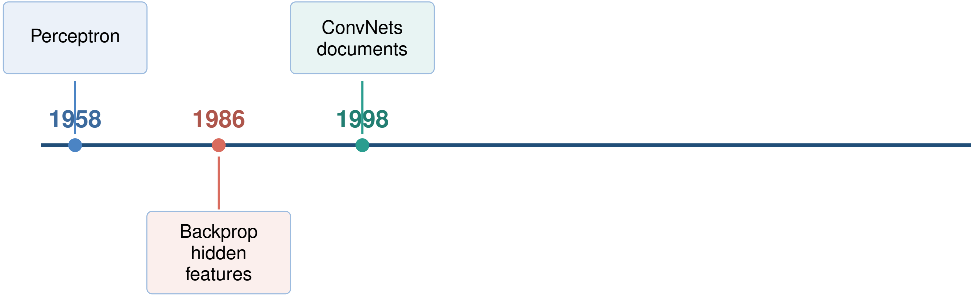
ML traditions overlap



Optional rapid check — reveal

Question: Is deep learning outside statistical modeling?
Answer: No. These communities and toolkits overlap rather than form a historical ladder; modern neural systems still use statistical objectives and inference.

The ideas are old; the operating regime changed



Sources: [S12] Rosenblatt, *Psychological Review*, 1958; [S01] Rumelhart, Hinton, and Williams, *Nature*, 1986; [S02] LeCun et al., *Proc. IEEE*, 1998; [S03] Krizhevsky, Sutskever, and Hinton, NeurIPS, 2012; [S04] Vaswani et al., NeurIPS, 2017; [S05] Bommasani et al., Stanford CRFM, 2021.

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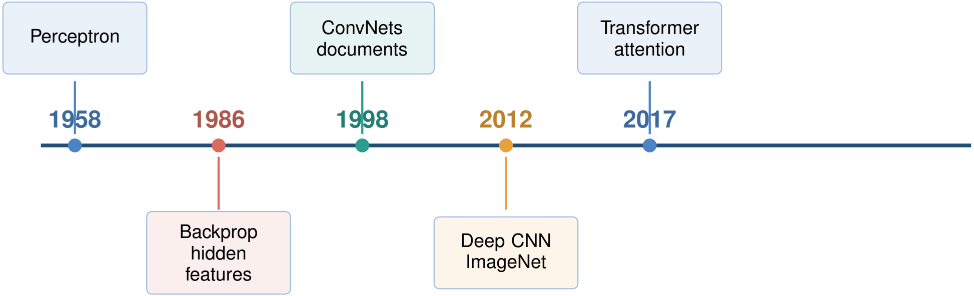

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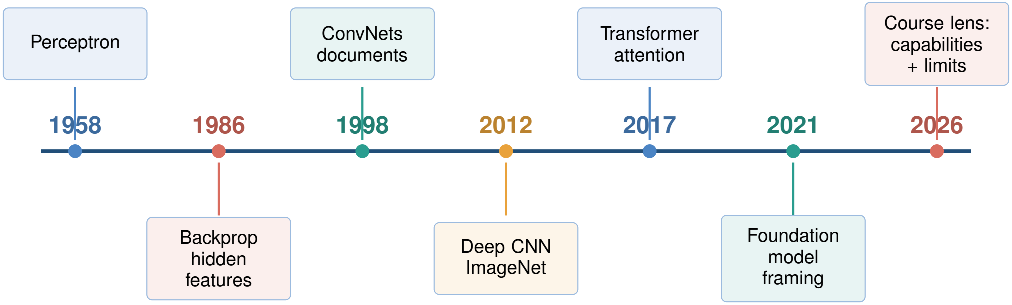

Reusable ingredients accumulated; data scale, parallel compute, software stacks, and broad adaptation changed what could be built together.

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Optional rapid check

Optional rapid check

What information from the forward pass might the backward pass need?

Commit to an answer before advancing the slide.

A deep model is a composition of parameterized maps

$$\mathbf{x} \xrightarrow{\mathbf{h}_0} f_1(\,\cdot\,;\boldsymbol{\theta}_1) \xrightarrow{\mathbf{h}_1} f_2(\,\cdot\,;\boldsymbol{\theta}_2) \rightarrow \cdots \xrightarrow{\mathbf{h}_{L-1}} f_L(\,\cdot\,;\boldsymbol{\theta}_L) \rightarrow \widehat{y} \rightarrow \boxed{\text{scalar loss } L \in \mathbb{R}}$$

$$f_{\boldsymbol{\theta}} = f_L \circ \cdots \circ f_2 \circ f_1, \qquad \boldsymbol{\theta} = (\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L).$$

$$\nabla_{\mathbf{h}_{\ell-1}} L = \mathbf{J}_{f_\ell}(\mathbf{h}_{\ell-1})^\top \nabla_{\mathbf{h}_\ell} L, \qquad \mathbf{J}_{f_\ell} \in \mathbb{R}^{d_\ell \times d_{\ell-1}}.$$

Optional rapid check — reveal

Question: What information from the forward pass might the backward pass need?
Answer: Intermediate activations and local derivatives—or recomputation of them—connect the local chain-rule factors while shapes match at every arrow.

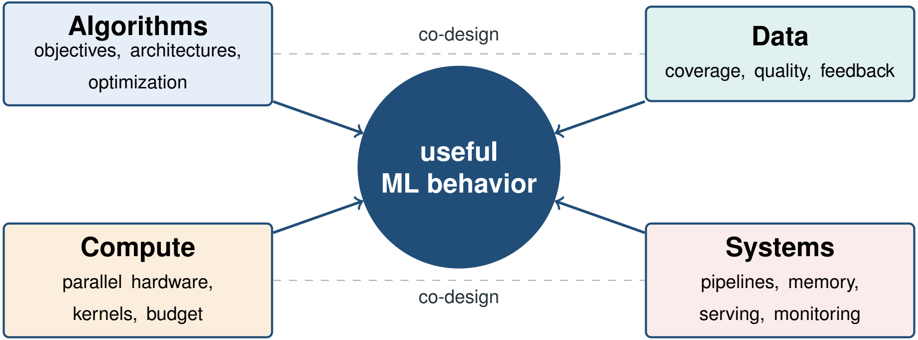
Think–Pair–Share

Think–Pair–Share

Rank the four factors—or choose the single factor that best explains modern progress.

Commit to an answer before advancing the slide.

Progress combines algorithms, data, compute, and systems



Think–Pair–Share — reveal

Question: Rank the four factors—or choose the single factor that best explains modern progress.
Answer: No single factor is sufficient. Progress comes from their interaction, and the bottleneck shifts by regime; more of one cannot automatically repair poor coverage, the wrong objective, an unreliable pipeline, or an infeasible budget.

Perception and generation now share multimodal models

Language
tokens $\leftrightarrow$ text, code,
structured output

broadly trained model
shared parameters
+ adaptation

Vision
pixels $\rightarrow$ labels,
regions, descriptions

Sources: [S05] Bommasani et al., Stanford CRFM, 2021; [S06] AI Index Steering Committee, Stanford HAI, 2026.

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Language
tokens ↔ text, code,
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Vision
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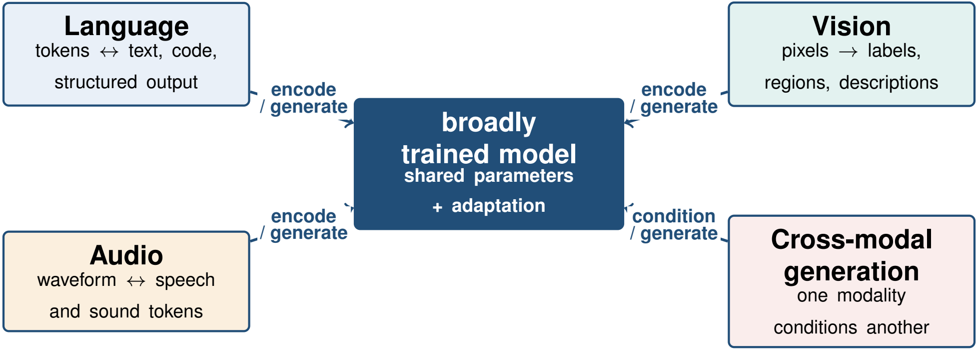
Audio
waveform ↔ speech
and sound tokens

Cross-modal generation
one modality
conditions another

Sources: [S05] Bommasani et al., Stanford CRFM, 2021; [S06] AI Index Steering Committee, Stanford HAI, 2026.

Perception and generation now share multimodal models



Shared interfaces enable many tasks; they do not imply equal competence across modalities, benchmarks, or deployment settings.

Sources: [S05] Bommasani et al., Stanford CRFM, 2021; [S06] AI Index Steering Committee, Stanford HAI, 2026.

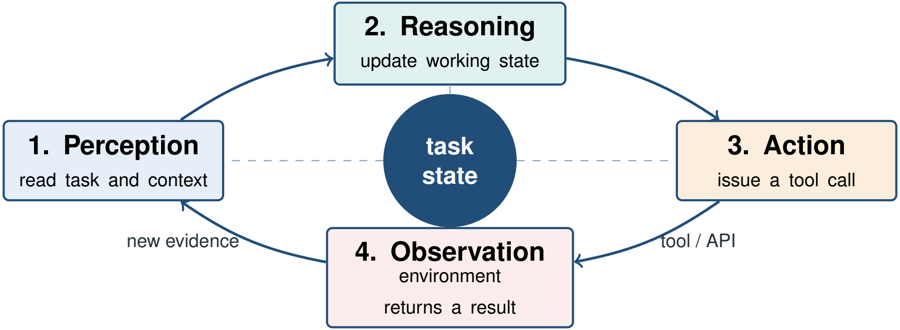
Pause & Predict

Pause & Predict

What turns a chatbot into an agent: a larger model, or a feedback loop with actions and observations?

Commit to an answer before advancing the slide.

Agents couple models to tools and environments



Pause & Predict — reveal

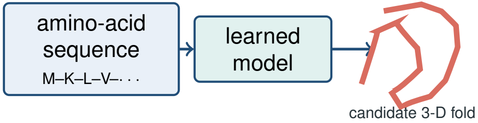
Question: What turns a chatbot into an agent: a larger model, or a feedback loop with actions and observations?
Answer: Agency here comes from coupling the model to tools and environment feedback; scale alone does not define it. ReAct interleaves reasoning, actions, and observations, while permissions, stopping, verification, and oversight remain system choices.

Sources: [S10] Yao et al., ICLR, 2023.

Machine learning is becoming a scientific instrument

Example: protein structure prediction

In CASP14, AlphaFold achieved accuracy competitive with experimental structures in a majority of cases and substantially exceeded other entered methods.



Sources: [S09] Jumper et al., *Nature*, 2021.

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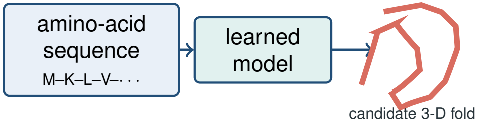
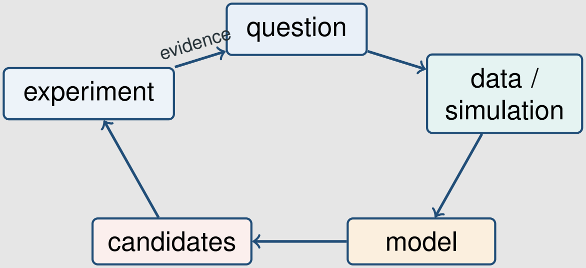


General discovery loop



ML can prioritize hypotheses or infer latent structure; scientific claims still depend on domain assumptions, uncertainty, and empirical validation.

Sources: [S09] Jumper et al., *Nature*, 2021.

Think–Pair–Share

Think–Pair–Share

A model exceeds a benchmark's human baseline. Is it deployment-ready?

Commit to an answer before advancing the slide.

Benchmark gains do not eliminate reliability gaps

Evaluation | **Distribution shift** | **Hallucination**

Robustness | **Safety** | **Deployment constraints**

GOVERN is cross-cutting: it informs and is infused throughout the other three functions.
MAP, MEASURE, and MANAGE may be performed in any order; the process should be iterative.

MAP
establish context to frame risks related to an AI system

MEASURE
analyze, assess, benchmark, and monitor AI risk and related impacts

MANAGE
allocate risk resources to mapped and measured risks

Think–Pair–Share — reveal

Question: A model exceeds a benchmark's human baseline. Is it deployment-ready?
Answer: Not necessarily. Benchmark scope, distribution shift, calibration, robustness, safety, and operating context still matter.

Sources: [S06] AI Index Steering Committee, Stanford HAI, 2026; [S11] Tabassi and NIST, AI RMF 1.0, 2023.

Pause & Predict

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With a fixed compute budget, should we keep increasing parameters while holding data fixed?

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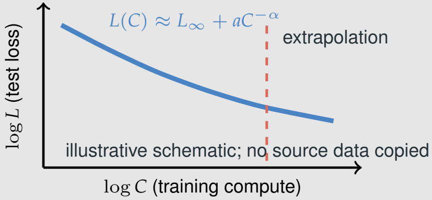
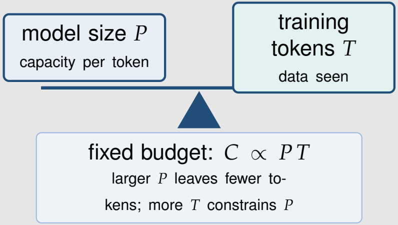
Scaling laws describe regimes, not guarantees

Empirical fit in a measured regime



Finite compute forces a tradeoff



- Power laws summarized loss trends in the language-model regimes studied by Kaplan et al.
- Hoffmann et al. found near-equal model/token scaling under their compute-optimal setup across more than 400 runs.
- A regime can change with data quality, architecture, objective, or evaluation; extrapolation is a hypothesis.

Pause & Predict — reveal

Question: With a fixed compute budget, should we keep increasing parameters while holding data fixed?
Answer: No. Compute-optimal training balances model size and training data, and the empirical relation is regime-dependent.

Sources: [S07] Kaplan et al., 2020; [S08] Hoffmann et al., 2022.

Next: review the reusable mathematical language

Review / bridge	Question it lets us ask	Where it returns
Data	What is measured and represented?	datasets + every model
Geometry	Which directions and structures matter?	linear algebra
Uncertainty	What should we believe given evidence?	$p(y \mid \mathbf{x})$ + probability
Optimization	How do parameters improve an objective?	calculus + optimization
Composition	How do simple maps build a deep model?	layers + chain rule

We review familiar mathematics briskly and add only the two-frame calculus bridge needed later.

Table of Contents

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Notation

Shape check

If $\mathbf{A} \in \mathbb{R}^{4\times 3}$ and $\mathbf{x} \in \mathbb{R}^3$, what is the shape of $\mathbf{A}\mathbf{x}$? Would $\mathbf{x}^\top \mathbf{A}$ be legal?

Commit to an answer before advancing the slide.

Notation

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- Sets by uppercase calligraphic letters (e.g., $\mathcal{S}$).

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Recall

Can you name the addition, scalar-multiplication, distributive, and identity axioms before they are revealed?

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6. $\alpha(\mathbf{x} + \mathbf{y}) = \alpha\mathbf{x} + \alpha\mathbf{y}, \quad \forall \alpha \in \mathbb{R} \qquad \forall \mathbf{x}, \mathbf{y} \in \mathcal{V}$ — distributive

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4. $\forall \mathbf{x} \in \mathcal{V} \quad \exists (-\mathbf{x}) \in \mathcal{V}$ such that $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$ — $-\mathbf{x}$ being additive inverse
5. $(\alpha\beta)\mathbf{x} = \alpha(\beta\mathbf{x}), \quad \alpha, \beta \in \mathbb{R} \qquad \forall \mathbf{x} \in \mathcal{V}$ — associative under scalar multiplication
6. $\alpha(\mathbf{x} + \mathbf{y}) = \alpha\mathbf{x} + \alpha\mathbf{y}, \quad \forall \alpha \in \mathbb{R} \qquad \forall \mathbf{x}, \mathbf{y} \in \mathcal{V}$ — distributive
7. $(\alpha + \beta)\mathbf{x} = \alpha\mathbf{x} + \beta\mathbf{x}, \quad \forall \alpha, \beta \in \mathbb{R} \qquad \forall \mathbf{x} \in \mathcal{V}$ — distributive

Vector spaces

Convention. We work over $\mathbb{R}$; most results extend to the complex field $\mathbb{C}$.
A vector space or *linear space* (over the field $\mathbb{R}$) consists of

(a) a set of vectors $\mathcal{V}$

(b) an addition operation: $\mathcal{V} \times \mathcal{V} \to \mathcal{V}$

(c) a scalar multiplication operation: $\mathbb{R} \times \mathcal{V} \to \mathcal{V}$

(d) a distinguished element $\mathbf{0} \in \mathcal{V}$

and satisfies the following properties:

1. $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}, \qquad \mathbf{x}, \mathbf{y} \in \mathcal{V}$ — commutative under addition
2. $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z}), \forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathcal{V}$ — associative under addition
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8. $1\mathbf{x} = \mathbf{x}, \forall \mathbf{x} \in \mathcal{V}$ — Scalar 1 being multiplicative identity

Vector spaces contd.

Example 1 (Vector space)

1. $\mathcal{V}_1 = \{\mathbf{0}\}$ for $\mathbf{0} \in \mathbb{R}^p$
2. $\mathcal{V}_2 = \mathbb{R}^p$
3. $\mathcal{V}_3 = \left\{\sum_{i=1}^k \alpha_i \mathbf{x}_i : \alpha_i \in \mathbb{R}\right\}$ for fixed $\mathbf{x}_i \in \mathbb{R}^p$

Vector spaces contd.

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Definition 2 (Subspace)

Vector spaces contd.

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Definition 2 (Subspace)

A **subspace** $\mathcal{S} \subseteq \mathcal{V}$ contains $\mathbf{0}$ and is closed under linear combinations: if $\mathbf{x}, \mathbf{y} \in \mathcal{S}$ and $\alpha, \beta \in \mathbb{R}$, then $\alpha\mathbf{x} + \beta\mathbf{y} \in \mathcal{S}$.

Vector spaces contd.

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Example 3 (Subspace)

$\mathcal{V}_1$, $\mathcal{V}_2$, and $\mathcal{V}_3$ in the example above are subspaces of $\mathbb{R}^p$.

Vector spaces contd.

Definition 4 (Span)

The **span** of a set of vectors, $\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$, is the set of all possible linear combinations of these vectors; i.e.,

$$\text{span}\{\mathbf{x}_1, \ldots, \mathbf{x}_k\} = \{\alpha_1\mathbf{x}_1 + \ldots + \alpha_k\mathbf{x}_k \,|\, \alpha_1, \ldots, \alpha_k \in \mathbb{R}\}. \tag{1}$$

Vector spaces contd.

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Definition 5 (Linear independence)

A set of vectors, $\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$, is **linearly independent** if

Vector spaces contd.

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Definition 5 (Linear independence)

A set of vectors, $\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$, is **linearly independent** if

$$\alpha_1 \mathbf{x}_1 + \ldots + \alpha_k \mathbf{x}_k = \mathbf{0} \Rightarrow \alpha_1 = \alpha_2 = \ldots = \alpha_k = 0\,. \tag{2}$$

Vector spaces contd.

Dimension check

Can three linearly independent vectors in $\mathbb{R}^5$ span all of $\mathbb{R}^5$? Why or why not?

Commit to an answer before advancing the slide.

Vector spaces contd.

Definition 6 (Basis)

The **basis** of a vector space, $\mathcal{V}$, is a set of vectors $\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$ that satisfy

1. $\mathcal{V} = \text{span}\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$,
2. $\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$ are linearly independent.

Vector spaces contd.

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Definition 7 (Dimension)

Every basis of a finite-dimensional vector space $\mathcal{V}$ has the same number of vectors. This common number is the **dimension** of $\mathcal{V}$, denoted $\dim(\mathcal{V})$.

Vector Norms

Definition 8 (Vector norm)

A norm of a vector in $\mathbb{R}^p$ is a function $||\cdot|| : \mathbb{R}^p \to \mathbb{R}$ such that for all vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^p$ and scalar $\lambda \in \mathbb{R}$

Recall

Which four properties distinguish a norm from an arbitrary size function?

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(a) $||\mathbf{x}|| \geq 0$ for all $\mathbf{x} \in \mathbb{R}^p$ — non-negativity

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- There is a family of ℓ_q-norms parameterized by $q \in [1, \infty]$;
- For $1 \leq q < \infty$, $\|\mathbf{x}\|_q := \left(\sum_{i=1}^p |x_i|^q\right)^{1/q}$; the endpoint is defined separately by $\|\mathbf{x}\|_\infty := \max_i |x_i|$.

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Example 9

1. ℓ_2-norm: $\|\mathbf{x}\|_2 := \sqrt{\sum_{i=1}^p x_i^2}$ — Euclidean norm

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3. ℓ_∞-norm: $||\mathbf{x}||_\infty := \max_{i=1,\ldots,p} |x_i|$ — Chebyshev norm

Vector norms contd.

Definition 10 (Quasi-norm)

A quasi-norm satisfies all the norm properties except (d) triangle inequality, which is replaced by $\|\mathbf{x} + \mathbf{y}\| \leq c\,(\|\mathbf{x}\| + \|\mathbf{y}\|)$ for a constant $c \geq 1$.

Vector norms contd.

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Definition 11 (Semi(pseudo)-norm)

A semi(pseudo)-norm satisfies all the norm properties except (b) definitiveness.

Vector norms contd.

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Definition 12 (ℓ_0-"norm")

$$\|\mathbf{x}\|_0 := \lim_{q \downarrow 0} \sum_{i=1}^{p} |x_i|^q = |\{i : x_i \neq 0\}|$$

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The ℓ_0-norm counts the non-zero components of $\mathbf{x}$. It is not a norm—for nonzero λ, $\|\lambda\mathbf{x}\|_0 = \|\mathbf{x}\|_0$ in general, so homogeneity fails. Hence it is also neither a **quasi-** nor a **semi-norm** under the definitions above.

Vector norms contd.

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Definition 13 (Norm balls)

Radius r in ℓ_q-norm: $\mathcal{B}_q(r) = \{\mathbf{x} \in \mathbb{R}^p : \|\mathbf{x}\|_q \leq r\}$.

Inner products

Definition 14 (Inner product)

The **inner product** of any two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^p$ (denoted by $\langle \cdot, \cdot \rangle$) is defined as

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3. $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$ and $\langle \mathbf{x}, \mathbf{x} \rangle = 0 \Leftrightarrow \mathbf{x} = \mathbf{0}$, $\forall \mathbf{x} \in \mathbb{R}^p$ positive definiteness

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Important relations involving the inner product:

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Important relations involving the inner product:

- Hölder's inequality: $|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\|_q \, \|\mathbf{y}\|_{q^*}$, where $q, q^* \in [1, \infty]$ and $\frac{1}{q} + \frac{1}{q^*} = 1$ (including $1^* = \infty$).

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3. $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$ and $\langle \mathbf{x}, \mathbf{x} \rangle = 0 \Leftrightarrow \mathbf{x} = \mathbf{0}$, $\forall \mathbf{x} \in \mathbb{R}^p$ positive definiteness

Important relations involving the inner product:

- Hölder's inequality: $|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\|_q \|\mathbf{y}\|_{q^*}$, where $q, q^* \in [1, \infty]$ and $\frac{1}{q} + \frac{1}{q^*} = 1$ (including $1^* = \infty$).
- Cauchy–Schwarz is the special case $q = q^* = 2$ of Hölder's inequality.

Vector norms contd.

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$$||\mathbf{x}||^\star = \sup_{||\mathbf{y}|| \leq 1} \left| \mathbf{x}^\top \mathbf{y} \right|, \qquad \mathbf{x} \in \mathbb{R}^p.$$

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Example 16

1. ℓ_2 is self-dual: $\sup_{\|\mathbf{x}\|_2\leq 1} \left|\mathbf{z}^\top \mathbf{x}\right| = \|\mathbf{z}\|_2$.
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Check: the constraint uses the primal norm; the supremum produces its dual.

Metrics

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Let $\mathcal{X}$ be a set, then a function $d(\cdot, \cdot) : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is a metric if $\forall \mathbf{x}, \mathbf{y} \in \mathcal{X}$:

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- **Norm** induce metrics while **pseudo-norms** induce pseudo-metrics

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Basic matrix definitions contd.

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Definition 22 (Recall the definition of span)

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  $\Rightarrow$ In that basis case, any vector $\mathbf{v} \in \mathbb{R}^n$ can be written uniquely as a linear combination of $\mathbf{x}_1$ through $\mathbf{x}_k$.

- The **projection** of a vector $\mathbf{y} \in \mathbb{R}^n$ onto the span of $\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$, is the vector $\mathbf{v} \in$ span$(\{\mathbf{x}_1, \ldots, \mathbf{x}_k\})$ such that $\mathbf{v}$ is as close as possible to $\mathbf{y}$:

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$$\text{Proj}_{\text{span}\{\mathbf{x}_1,\ldots,\mathbf{x}_k\}}(\mathbf{y}) = \underset{\mathbf{v} \in \text{span}\{\mathbf{x}_1,\ldots,\mathbf{x}_k\}}{\arg\min} \|\mathbf{y} - \mathbf{v}\|_2 \tag{8}$$

Basic matrix definitions contd.

Projection check

When is $\mathbf{A}(\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top\mathbf{y}$ a valid projection formula, and what should replace it otherwise?

Commit to an answer before advancing the slide.

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The **range** of a matrix, $\mathbf{A} \in \mathbb{R}^{n \times p}$, (denoted by $\mathrm{range}(\mathbf{A})$) is defined as

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- The projection of a vector $\mathbf{y} \in \mathbb{R}^n$ onto the range of $\mathbf{A}$ is given by:

$$\mathrm{Proj}_{\mathrm{range}(\mathbf{A})}(\mathbf{y}) = \underset{\mathbf{v} \in \mathrm{range}(\mathbf{A})}{\arg\min} \|\mathbf{v} - \mathbf{y}\|_2 = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1}\mathbf{A}^\top \mathbf{y}\,, \tag{10}$$

where $\mathbf{A}$ has full column rank, so $p \leq n$.

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- $\operatorname{range}(\mathbf{A})$ is the span of the columns (or the **column space**) of $\mathbf{A}$.
- The projection of a vector $\mathbf{y} \in \mathbb{R}^n$ onto the range of $\mathbf{A}$ is given by:

$$\operatorname{Proj}_{\operatorname{range}(\mathbf{A})}(\mathbf{y}) = \underset{\mathbf{v} \in \operatorname{range}(\mathbf{A})}{\arg\min} \|\mathbf{v} - \mathbf{y}\|_2 = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{y}\,, \tag{10}$$

where $\mathbf{A}$ has full column rank, so $p \leq n$.

$\Rightarrow$ This is the fitted vector for least squares. For arbitrary rank, the same unique projection is $\mathbf{A}\mathbf{A}^\dagger \mathbf{y}$.

Basic matrix definitions contd.

Definition 24 (Nullspace of a matrix)

The **nullspace** of a matrix, $\mathbf{A} \in \mathbb{R}^{n \times p}$, (denoted by $\text{null}(\mathbf{A})$) is defined as

$$\text{null}(\mathbf{A}) = \{\mathbf{x} \in \mathbb{R}^p \mid \mathbf{A}\mathbf{x} = \mathbf{0}\} \tag{11}$$

Basic matrix definitions contd.

Definition 24 (Nullspace of a matrix)

The **nullspace** of a matrix, $\mathbf{A} \in \mathbb{R}^{n \times p}$, (denoted by $\mathrm{null}(\mathbf{A})$) is defined as

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- $\mathrm{null}(\mathbf{A})$ is the set of vectors mapped to zero by $\mathbf{A}$.

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- $\mathrm{null}(\mathbf{A})$ is the set of vectors mapped to zero by $\mathbf{A}$.
- $\mathrm{null}(\mathbf{A})$ is the set of vectors orthogonal to the *rows* of $\mathbf{A}$.

Basic matrix definitions contd.

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- $\mathrm{null}(\mathbf{A})$ is the set of vectors mapped to zero by $\mathbf{A}$.
- $\mathrm{null}(\mathbf{A})$ is the set of vectors orthogonal to the *rows* of $\mathbf{A}$.
- The vectors in $\mathrm{range}(\mathbf{A})$ are in $\mathbb{R}^n$, while the vectors in $\mathrm{null}(\mathbf{A})$ are in $\mathbb{R}^p$.

Basic matrix definitions contd.

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- The vectors in $\operatorname{range}(\mathbf{A})$ are in $\mathbb{R}^n$, while the vectors in $\operatorname{null}(\mathbf{A})$ are in $\mathbb{R}^p$.
  $\Rightarrow$ vectors in $\operatorname{range}(\mathbf{A}^\top)$ and $\operatorname{null}(\mathbf{A})$ are both in $\mathbb{R}^p$.

Basic matrix definitions contd.

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- The vectors in $\mathrm{range}(\mathbf{A})$ are in $\mathbb{R}^n$, while the vectors in $\mathrm{null}(\mathbf{A})$ are in $\mathbb{R}^p$.
  $\Rightarrow$ vectors in $\mathrm{range}(\mathbf{A}^\top)$ and $\mathrm{null}(\mathbf{A})$ are both in $\mathbb{R}^p$.
- $\mathrm{range}(\mathbf{A}^\top)$ and $\mathrm{null}(\mathbf{A})$ intersect only at $\mathbf{0}$ and together span $\mathbb{R}^p$.

Basic matrix definitions contd.

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  $\Rightarrow$ vectors in $\mathrm{range}(\mathbf{A}^\top)$ and $\mathrm{null}(\mathbf{A})$ are both in $\mathbb{R}^p$.
- $\mathrm{range}(\mathbf{A}^\top)$ and $\mathrm{null}(\mathbf{A})$ intersect only at $\mathbf{0}$ and together span $\mathbb{R}^p$.
  These subspaces are **orthogonal complements**: $\mathrm{range}(\mathbf{A}^\top) = \mathrm{null}(\mathbf{A})^\perp$.

Basic matrix definitions contd.

Definition 25 (Rank of a matrix)

The **rank** of a matrix, $\mathbf{A} \in \mathbb{R}^{n \times p}$, (denoted by $\mathrm{rank}(\mathbf{A})$) is defined as

$$\mathrm{rank}(\mathbf{A}) = \dim(\mathrm{range}(\mathbf{A})) \tag{12}$$

Basic matrix definitions contd.

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- $\mathrm{rank}(\mathbf{A})$ is the maximum number of independent columns (or rows) of $\mathbf{A}$,

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- $\mathrm{rank}(\mathbf{A})$ is the maximum number of independent columns (or rows) of $\mathbf{A}$,
  $\Rightarrow \mathrm{rank}(\mathbf{A}) \leq \min(n, p)$.

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Basic matrix definitions contd.

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- $\mathrm{rank}(\mathbf{A}) = \mathrm{rank}(\mathbf{A}^\top)$; and $\mathrm{rank}(\mathbf{A}) + \dim(\mathrm{null}(\mathbf{A})) = p$.

Basic matrix definitions contd.

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- $\operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{A}^\top)$; and $\operatorname{rank}(\mathbf{A}) + \dim(\operatorname{null}(\mathbf{A})) = p$.
- For $\mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{B} \in \mathbb{R}^{n \times p}, \operatorname{rank}(\mathbf{AB}) \leq \min(\operatorname{rank}(\mathbf{A}), \operatorname{rank}(\mathbf{B}))$

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- For $\mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{B} \in \mathbb{R}^{n \times p}, \operatorname{rank}(\mathbf{AB}) \leq \min(\operatorname{rank}(\mathbf{A}), \operatorname{rank}(\mathbf{B}))$
- For $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}$, $\operatorname{rank}(\mathbf{A} + \mathbf{B}) \leq \operatorname{rank}(\mathbf{A}) + \operatorname{rank}(\mathbf{B})$

Basic matrix definitions contd.

Definition 26 (The inverse)

The **inverse** of a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is denoted $\mathbf{A}^{-1}$, and is the unique matrix such that

Basic matrix definitions contd.

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Basic matrix definitions contd.

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The **inverse** of a square matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ is denoted $\mathbf{A}^{-1}$, and is the unique matrix such that

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- Not all matrices have inverses. Non-square matrices do not have inverses by definition.

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- A square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ has an inverse if and only if $\mathrm{rank}(\mathbf{A}) = n$.

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- The following are properties of the inverse (all assume that $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ are non-singular):

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  - $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$
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  - $(\mathbf{A}^{-1})^{\top} = (\mathbf{A}^{\top})^{-1}$

Basic matrix definitions contd.

Definition 27 (Orthogonal matrices)

- Two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are **orthogonal** if $\mathbf{x}^\top \mathbf{y} = 0$.

Basic matrix definitions contd.

Definition 27 (Orthogonal matrices)

- Two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are **orthogonal** if $\mathbf{x}^\top \mathbf{y} = 0$.
- A vector $\mathbf{x} \in \mathbb{R}^n$ is **normalized** if $\|\mathbf{x}\|_2 = 1$.

Basic matrix definitions contd.

Definition 27 (Orthogonal matrices)

- Two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are **orthogonal** if $\mathbf{x}^\top \mathbf{y} = 0$.
- A vector $\mathbf{x} \in \mathbb{R}^n$ is **normalized** if $\|\mathbf{x}\|_2 = 1$.
- A square matrix $\mathbf{U} \in \mathbb{R}^{n \times n}$ is **orthogonal** if all its columns are orthogonal to each other and are normalized (the columns are then referred to as being **orthonormal**):

$$\mathbf{U}^\top \mathbf{U} = \mathbf{I}_n = \mathbf{U}\mathbf{U}^\top \tag{14}$$

Basic matrix definitions contd.

Definition 27 (Orthogonal matrices)

- Two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are **orthogonal** if $\mathbf{x}^\top \mathbf{y} = 0$.
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- The inverse of an orthogonal matrix is its transpose: $\mathbf{U}^{-1} = \mathbf{U}^\top$

Basic matrix definitions contd.

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- The inverse of an orthogonal matrix is its transpose: $\mathbf{U}^{-1} = \mathbf{U}^\top$
- Operating on a vector with an orthogonal matrix will not change its Euclidean norm, i.e.,

$$\|\mathbf{U}\mathbf{x}\|_2 = \|\mathbf{x}\|_2 \ , \tag{15}$$

for any $\mathbf{x} \in \mathbb{R}^n, \mathbf{U} \in \mathbb{R}^{n \times n}$ orthogonal.

Basic matrix definitions contd.

Definition 27 (Orthogonal matrices)

- Two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are **orthogonal** if $\mathbf{x}^\top \mathbf{y} = 0$.
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Definition 28 (Symmetric matrix)

A matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ is **symmetric** if $\mathbf{A} = \mathbf{A}^\top$. It is **skew-symmetric** (or anti-symmetric) if $\mathbf{A} = -\mathbf{A}^\top$.

Basic matrix definitions contd.

PSD check

Why is $\mathbf{B}^\top\mathbf{B}$ positive semidefinite for every real matrix $\mathbf{B}$? Give a one-line certificate.

Commit to an answer before advancing the slide.

Basic matrix definitions contd.

Definition 29 (Positive semi-definite & positive definite matrices)

A symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is **positive semi-definite** (denoted $\mathbf{A} \succeq 0$) if $\mathbf{x}^\top \mathbf{A}\mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$; while it is **positive definite** (denoted $\mathbf{A} \succ 0$) if $\mathbf{x}^\top \mathbf{A}\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

Basic matrix definitions contd.

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- $\mathbf{B}^\top \mathbf{B} \succeq 0$ because $\mathbf{x}^\top \mathbf{B}^\top \mathbf{B}\mathbf{x} = \|\mathbf{B}\mathbf{x}\|_2^2 \geq 0$ for every $\mathbf{x}$.
- $\mathbf{A} \succeq 0$ iff all its eigenvalues are **non-negative**, i.e., $\lambda_{\min}(\mathbf{A}) \geq 0$.

Basic matrix definitions contd.

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- $\mathbf{A} \succeq 0$ iff all its eigenvalues are **non-negative**, i.e., $\lambda_{\min}(\mathbf{A}) \geq 0$.
- Similarly, $\mathbf{A} \succ 0$ iff all its eigenvalues are **positive**, i.e., $\lambda_{\min}(\mathbf{A}) > 0$.

Basic matrix definitions contd.

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- $\mathbf{B}^\top \mathbf{B} \succeq 0$ because $\mathbf{x}^\top \mathbf{B}^\top \mathbf{B}\mathbf{x} = \|\mathbf{B}\mathbf{x}\|_2^2 \geq 0$ for every $\mathbf{x}$.
- $\mathbf{A} \succeq 0$ iff all its eigenvalues are **non-negative**, i.e., $\lambda_{\min}(\mathbf{A}) \geq 0$.
- Similarly, $\mathbf{A} \succ 0$ iff all its eigenvalues are **positive**, i.e., $\lambda_{\min}(\mathbf{A}) > 0$.
- $\mathbf{A}$ is negative semi-definite if $-\mathbf{A} \succeq 0$; while $\mathbf{A}$ is negative definite if $-\mathbf{A} \succ 0$.

Basic matrix definitions contd.

Definition 29 (Positive semi-definite & positive definite matrices)

A symmetric matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ is **positive semi-definite** (denoted $\mathbf{A} \succeq 0$) if $\mathbf{x}^\top \mathbf{A}\mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$; while it is **positive definite** (denoted $\mathbf{A} \succ 0$) if $\mathbf{x}^\top \mathbf{A}\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

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- $\mathbf{A} \succeq 0$ iff all its eigenvalues are **non-negative**, i.e., $\lambda_{\min}(\mathbf{A}) \geq 0$.
- Similarly, $\mathbf{A} \succ 0$ iff all its eigenvalues are **positive**, i.e., $\lambda_{\min}(\mathbf{A}) > 0$.
- $\mathbf{A}$ is negative semi-definite if $-\mathbf{A} \succeq 0$; while $\mathbf{A}$ is negative definite if $-\mathbf{A} \succ 0$.
- Semi-definite ordering of two *symmetric* matrices, $\mathbf{A}$ and $\mathbf{B}$: $\mathbf{A} \succeq \mathbf{B}$ if $\mathbf{A} - \mathbf{B} \succeq 0$.

Basic matrix definitions contd.

Definition 29 (Positive semi-definite & positive definite matrices)

A symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is **positive semi-definite** (denoted $\mathbf{A} \succeq 0$) if $\mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$; while it is **positive definite** (denoted $\mathbf{A} \succ 0$) if $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

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- $\mathbf{A} \succeq 0$ iff all its eigenvalues are **non-negative**, i.e., $\lambda_{\min}(\mathbf{A}) \geq 0$.
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- Semi-definite ordering of two *symmetric* matrices, $\mathbf{A}$ and $\mathbf{B}$: $\mathbf{A} \succeq \mathbf{B}$ if $\mathbf{A} - \mathbf{B} \succeq 0$.

Example 30 (Matrix inequalities)

Basic matrix definitions contd.

Definition 29 (Positive semi-definite & positive definite matrices)

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1. If $\mathbf{A} \succeq 0$ and $\mathbf{B} \succeq 0$, then $\mathbf{A} + \mathbf{B} \succeq 0$

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6. If $\mathbf{A} \succ 0$, then $\mathbf{A}^{-1} \succ 0$

Matrix decompositions

Definition 31 (Eigenvalues & Eigenvectors)

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A nonzero vector $\mathbf{x}$ is an **eigenvector** of a *square* matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ if $\mathbf{Ax} = \lambda\mathbf{x}$. For a general real matrix, λ and $\mathbf{x}$ may be complex; for real symmetric $\mathbf{A}$, they can be chosen real.

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- *The trace of $\mathbf{A}$ is equal to the sum of its eigenvalues, trace$(\mathbf{A}) = \sum_{i=1}^{n} \lambda_i$*

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- *If $\mathbf{A}$ is non-singular, then $1/\lambda_i$ is an eigenvalue of $\mathbf{A}^{-1}$ with associated eigenvector $\mathbf{x}_i$, i.e., $\mathbf{A}^{-1}\mathbf{x}_i = (1/\lambda_i)\mathbf{x}_i$*

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- *The eigenvalues of a diagonal matrix $\mathbf{D} =$ diag$(d_1, \ldots, d_n)$ are just the diagonal entries $d_1, \ldots, d_n$*

Matrix decompositions contd.

Definition 33 (Eigenvalue decomposition)

If a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ has a basis of real eigenvectors, its **eigenvalue decomposition** is

$$\mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1} \tag{16}$$

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If $\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric, the decomposition becomes $\mathbf{A} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$, where $\mathbf{U} \in \mathbb{R}^{n \times n}$ is orthogonal, i.e., $\mathbf{U}^\top\mathbf{U} = \mathbf{I}_n$ and λ_i are real.

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If we order $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n$, $\lambda_i(\mathbf{A})$ becomes the i^{th} largest eigenvalue of $\mathbf{A}$:

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If we order $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n$, $\lambda_i(\mathbf{A})$ becomes the i^{th} largest eigenvalue of $\mathbf{A}$:

- $\lambda_n(\mathbf{A}) = \lambda_{\text{min}}(\mathbf{A})$ is the minimum eigenvalue of $\mathbf{A}$
- $\lambda_1(\mathbf{A}) = \lambda_{\text{max}}(\mathbf{A})$ is the maximum eigenvalue of $\mathbf{A}$

Matrix decompositions contd

Thin-SVD shape check

If $\mathbf{A} \in \mathbb{R}^{n \times p}$ has rank r, what are the shapes of $\mathbf{U}$, $\mathbf{\Sigma}$, and $\mathbf{V}$ in its thin SVD?

Commit to an answer before advancing the slide.

Matrix decompositions contd

Definition 35 (Singular value decomposition)

The **singular value decomposition** (SVD) of a matrix, $\mathbf{A} \in \mathbb{R}^{n \times p}$, is given by:

$$\mathbf{A} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \tag{17}$$

- $\mathrm{rank}(\mathbf{A}) = r \leq \min(n, p)$ and σ_i is the i^{th} **singular value** of $\mathbf{A}$

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- $\mathbf{u}_i$ and $\mathbf{v}_i$ are the i^{th} **left** and **right singular vectors** of $\mathbf{A}$ respectively
- $\mathbf{U} \in \mathbb{R}^{n \times r}$ and $\mathbf{V} \in \mathbb{R}^{p \times r}$ have orthonormal columns: $\mathbf{U}^\top\mathbf{U} = \mathbf{I}_r = \mathbf{V}^\top\mathbf{V}$

Matrix decompositions contd

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- $\mathbf{\Sigma} = \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$, where $\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_r > 0$ (for $r = 0$, the sum is empty)

Matrix decompositions contd

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- $\boldsymbol{\Sigma} = \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$, where $\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_r > 0$ (for $r = 0$, the sum is empty)
- $\mathbf{v}_i$ are eigenvectors of $\mathbf{A}^\top\mathbf{A}$, $\sigma_i = \sqrt{\lambda_i(\mathbf{A}^\top\mathbf{A})}$, and $\lambda_i(\mathbf{A}^\top\mathbf{A}) = 0$ for $i > r$, since $\mathbf{A}^\top\mathbf{A} = (\mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top)^\top(\mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top) = (\mathbf{V}\boldsymbol{\Sigma}^2\mathbf{V}^\top)$

Matrix decompositions contd

Definition 35 (Singular value decomposition)

The **singular value decomposition** (SVD) of a matrix, $\mathbf{A} \in \mathbb{R}^{n \times p}$, is given by:

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \tag{17}$$

- $\operatorname{rank}(\mathbf{A}) = r \leq \min(n, p)$ and σ_i is the i^{th} **singular value** of $\mathbf{A}$
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- $\mathbf{U} \in \mathbb{R}^{n \times r}$ and $\mathbf{V} \in \mathbb{R}^{p \times r}$ have orthonormal columns: $\mathbf{U}^\top\mathbf{U} = \mathbf{I}_r = \mathbf{V}^\top\mathbf{V}$
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Matrix decompositions contd

Definition 36 (LU)

With row pivoting, the **LU factorization** of a non-singular square matrix, $\mathbf{A} \in \mathbb{R}^{p \times p}$, is given by:

$$\mathbf{PA} = \mathbf{LU}\,, \tag{18}$$

where $\mathbf{P}$ is a **permutation matrix**[a], $\mathbf{L}$ is **lower triangular** and $\mathbf{U}$ is **upper triangular**.

[a] A matrix $\mathbf{P} \in \mathbb{R}^{p \times p}$ is **permutation** if it has only one 1 in each row and each column.

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For $n \geq p$ and full-column-rank $\mathbf{A} \in \mathbb{R}^{n \times p}$, the reduced **QR factorization** is $\mathbf{A} = \mathbf{QR}$, where $\mathbf{Q} \in \mathbb{R}^{n \times p}$ has orthonormal columns ($\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$), and $\mathbf{R} \in \mathbb{R}^{p \times p}$ is upper triangular.

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Definition 38 (Cholesky)

The **Cholesky factorization** of a positive definite and symmetric matrix, $\mathbf{A} \in \mathbb{R}^{p \times p}$, is given by $\mathbf{A} = \mathbf{LL}^\top$, where $\mathbf{L}$ is a **lower triangular** matrix with positive entries on the *diagonal*.

Matrix norms

1. Recall the norm contract

For $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times p}$ and $\lambda \in \mathbb{R}$, a matrix norm satisfies

$\|\mathbf{A}\| \geq 0$, with equality iff $\mathbf{A} = \mathbf{0}$;

$\|\mathbf{A} + \mathbf{B}\| \leq \|\mathbf{A}\| + \|\mathbf{B}\|$;

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the domain is $\mathbb{R}^{n \times p}$.

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2. Frobenius: entrywise size

$$\langle \mathbf{A}, \mathbf{B} \rangle_F = \operatorname{trace}(\mathbf{A}^\top \mathbf{B}),$$

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^{n} \sum_{j=1}^{p} A_{ij}^2} = \sqrt{\operatorname{trace}(\mathbf{A}^\top \mathbf{A})}.$$

3. Operator: largest stretch

For $q, r \in [1, \infty]$,

$$\|\mathbf{A}\|_{q \to r} = \sup_{\|\mathbf{x}\|_q \leq 1} \|\mathbf{A}\mathbf{x}\|_r \, .$$

In particular, $\|\mathbf{A}\|_{2 \to 2} = \sigma_1(\mathbf{A})$, with $\sigma_1(\mathbf{0}) := 0$.

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In particular, $\|\mathbf{A}\|_{2\to 2} = \sigma_1(\mathbf{A})$, with $\sigma_1(\mathbf{0}) := 0$.

Quick distinction

Frobenius measures all entries together; an operator norm asks for the largest input–output amplification.

Where linear algebra returns in this course

Reusable toolkit
shape contracts • norms
projections • factorizations

Where linear algebra returns in this course

Regression
$\mathbf{X} \in \mathbb{R}^{N\times D}$, $\widehat{\mathbf{y}} = \mathbf{X}\mathbf{w} \in \mathbb{R}^N$

Regularization
under suitable objectives,
$\|\mathbf{w}\|_1$ promotes sparsity;
$\|\mathbf{w}\|_2^2$ shrinks

PCA / compression
centered $\mathbf{X}$:
$\mathbf{X} \approx \mathbf{U}_k \boldsymbol{\Sigma}_k \mathbf{V}_k^\top$

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Covariance
centered $\mathbf{X}$: $\mathbf{C} = \mathbf{X}^\top \mathbf{X}/N \succeq 0$

Neural layers
$\mathbf{H} : B \times d$, $\mathbf{W} : d \times h$, $\mathbf{1} \in \mathbb{R}^B$, $\mathbf{b} \in \mathbb{R}^h$
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The reusable habit

Name every object's space, check every inner dimension, then ask what geometry the operation preserves or discards.

Calculus recap 1/2: Gradients

Recall before the bridge

For a differentiable $f : \mathbb{R}^D \to \mathbb{R}$, which unit direction gives the largest local decrease, and what qualification belongs in that claim?

Commit to an answer before advancing the slide.

Calculus recap 1/2: Gradients

Bridge to later lectures: notation we will reuse in optimization and neural networks.

1. Recall the sensitivities

In one variable,

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

For $\mathbf{x} = (x_1, \ldots, x_D)^\top$ and coordinate vector $\mathbf{e}_i$,

$$\frac{\partial f}{\partial x_i}(\mathbf{x}) = \lim_{h \to 0} \frac{f(\mathbf{x} + h\mathbf{e}_i) - f(\mathbf{x})}{h}.$$

The course uses a column gradient:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \partial f / \partial x_1 & \cdots & \partial f / \partial x_D \end{bmatrix}^\top \in \mathbb{R}^{D \times 1}.$$

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2. Use the local linear model

If f is differentiable at $\mathbf{x}$, then

$$f(\mathbf{x} + \Delta\mathbf{x}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \Delta\mathbf{x} + o(\|\Delta\mathbf{x}\|_2).$$

Thus for $\|\mathbf{d}\|_2 = 1$, the directional derivative is

$$D_{\mathbf{d}} f(\mathbf{x}) = \nabla f(\mathbf{x})^\top \mathbf{d}, \qquad (1 \times D)(D \times 1) = 1 \times 1.$$

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3. Result and quick check

Cauchy–Schwarz gives $|\nabla f(\mathbf{x})^\top \mathbf{d}| \le \|\nabla f(\mathbf{x})\|_2$. If $\nabla f(\mathbf{x}) \neq \mathbf{0}$, the locally steepest Euclidean descent direction is

$$\boxed{\mathbf{d}_- = -\frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|_2}}.$$

This is a **first-order, local** statement; a finite update still needs a step size. If $\nabla f(\mathbf{x}) = \mathbf{0}$, first order selects no direction.

Calculus recap 2/2: Chain rule and backpropagation

Shape check before the bridge

Let $\mathbf{g} : \mathbb{R}^D \to \mathbb{R}^K$ and $f : \mathbb{R}^K \to \mathbb{R}$. Which product maps the upstream K-gradient back to a gradient in $\mathbb{R}^D$?

Commit to an answer before advancing the slide.

Calculus recap 2/2: Chain rule and backpropagation

Bridge to later lectures: composition notation for the later backpropagation lecture.

1. Name the local map and its shape

The scalar rule is

$$(f \circ g)'(x) = f'(g(x))g'(x).$$

For $\mathbf{g} : \mathbb{R}^D \to \mathbb{R}^K$, define

$$[\mathbf{J}_{\mathbf{g}}(\mathbf{x})]_{ki} = \frac{\partial g_k}{\partial x_i}(\mathbf{x}), \qquad \mathbf{J}_{\mathbf{g}}(\mathbf{x}) \in \mathbb{R}^{K \times D}.$$

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2. Compose compatible shapes

With column gradients,

$$\boxed{\nabla_\mathbf{x}(f \circ \mathbf{g})(\mathbf{x}) = \mathbf{J}_\mathbf{g}(\mathbf{x})^\top \nabla f(\mathbf{g}(\mathbf{x}))}.$$

The check is

$$(D \times K)(K \times 1) = D \times 1.$$

Hence $\mathbf{J}_\mathbf{g} \nabla f$ is not the chain-rule product under these conventions.

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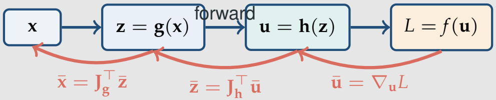
3. Preview reverse-mode backpropagation



For scalar L, $\bar{\mathbf{v}} := \nabla_{\mathbf{v}} L$ has the same dimension as $\mathbf{v}$. Backpropagation stores or recomputes needed forward intermediates, then propagates local Jacobian-transpose–vector products in reverse and sums contributions at branches. **Later:** optimization will decide how to use the resulting gradients.

Table of Contents

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Sample space Ω

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- Two events are called independent if and only if $P(A \cap B) = P(A)P(B)$.

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- (Union Bound) $P(A \cup B) \leq P(A) + P(B)$
- $P(\Omega \backslash A) = 1 - P(A)$
- If $A_1, \ldots, A_k$ partition Ω, then $\sum_{i=1}^{k} P(A_i) = 1$ and, for any event B,

$$P(B) = \sum_{i:P(A_i)>0} P(B \mid A_i)P(A_i) \qquad \text{(law of total probability).}$$

Definition 42 (Conditional probability and independence)

Let B be an event with $P(B) > 0$. The conditional probability of A given B is

$$P(A|B) \triangleq \frac{P(A \cap B)}{P(B)} \,. \tag{20}$$

- $P(A|B)$ is the probability measure of the event A after observing the occurrence of B.
- Two events are called independent if and only if $P(A \cap B) = P(A)P(B)$.
- Diagnostic: Are disjoint positive-probability events independent?

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- Answer: No—they are dependent.

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Random Variables

Definition 43 (Random Variable)

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- *A random variable must make value statements into legitimate events.*
- *Concretely, for each interval $C \subseteq \mathbb{R}$ used here, $\{\omega : X(\omega) \in C\}$ must belong to $\mathcal{F}$.*

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The probability of the set associated with a random variable X taking on some specific value k is

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Cumulative distribution functions

The CDF records the distribution of a real-valued random variable.

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- $x \leq y \Rightarrow F_X(x) \leq F_X(y)$
- F_X is right-continuous, and $P(a < X \leq b) = F_X(b) - F_X(a)$.

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Let a discrete random variable (RV) X take values in a finite or countable support $\mathcal{X}$.

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For a continuous RV with a density, probabilities are areas under a non-negative function.

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A **Probability Density Function** (PDF) $f_X : \mathbb{R} \to [0, \infty)$ satisfies

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt\,. \tag{25}$$

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- Answer: Yes. The PDF height is not a point probability: for such a continuous law,

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  - $\int_{\mathcal{A}} f_X(x)\, dx = P(X \in \mathcal{A})$ for suitable value sets $\mathcal{A}$.
- For example, $X \sim \mathrm{Uniform}(0, \frac{1}{2})$ has density height 2 but total area 1.

Expectation

Definition 51 (Expectation of random variable)

Discrete random variable: Suppose that X has support $\mathcal{X}$, PMF p_X, and $g : \mathbb{R} \to \mathbb{R}$ is a function for which the expectation exists.

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- For a discrete random variable X, $\mathbb{E}[1\{X = k\}] = P(X = k)$
- Diagnostic: Does linearity require independence?
- Answer: No.

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Definition 52 (The variance of a random variable X)

If X has a finite second moment, its variance is

$$\mathrm{Var}(X) \triangleq \mathbb{E}\left[(X - \mathbb{E}\left[X\right])^2\right] = \mathbb{E}\left[X^2\right] - \mathbb{E}\left[X\right]^2 . \tag{29}$$

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- $X \sim \text{Bernoulli}(q)$ (where $0 \le q \le 1$): one if a coin with heads probability q comes up heads, zero otherwise.

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- $X \sim$ Geometric(q) (where $0 < q \leq 1$): the number of flips of a coin with heads probability q until the first heads.

Some common discrete random variables

- $X \sim \text{Bernoulli}(q)$ (where $0 \leq q \leq 1$): one if a coin with heads probability q comes up heads, zero otherwise.

$$p_X(x) = \begin{cases} q & \text{if } x = 1 \\ 1 - q & \text{if } x = 0 \end{cases} . \tag{30}$$

- $X \sim \text{Binomial}(n, q)$ (where $0 \leq q \leq 1$): the number of heads in n independent flips of a coin with heads probability q

$$p_X(x) = \binom{n}{x} q^x (1-q)^{n-x}, \quad x \in \{0, \ldots, n\} . \tag{31}$$

- $X \sim \text{Geometric}(q)$ (where $0 < q \leq 1$): the number of flips of a coin with heads probability q until the first heads.

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$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}. \tag{36}$$

Table of Contents

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The joint CDF and the CDFs of each variable separately are related by

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- If either argument tends to $-\infty$, then $F_{XY}(x,y) \to 0$.

Joint and marginal Probability Mass Functions (PMFs)

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If X and Y are discrete random variables with supports $\mathcal{X}$ and $\mathcal{Y}$, then the **joint PMF** $p_{XY} : \mathcal{X} \times \mathcal{Y} \to [0, 1]$ is

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We refer to $p_X(x)$ as the **marginal PMF** of X.

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Definition 57 (Joint PDF)

Let X and Y be two continuous random variables with joint distribution function F_{XY}. When their joint law has a density and F_{XY} is twice differentiable, the **joint PDF** is

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Definition 58 (Marginal PDF)

The marginal PDF of X is $f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, dy$.

Conditional distributions

Definition 59 (Conditional distribution)

Discrete case: for x with $p_X(x) > 0$, the conditional PMF of Y given $X = x$ is

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In either ratio, the denominator must be strictly positive.

Bayes's rule

Diagnostic: What denominator conditions must be checked before applying Bayes's rule?

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Definition 60 (Bayes's rule)

For **discrete random variables** X, Y, at values with $p_X(x) > 0$ (and $p_Y(y) > 0$ for the likelihood form),

$$p_{Y|X}(y|x) = \frac{p_{XY}(x,y)}{p_X(x)} = \frac{p_{X|Y}(x|y)p_Y(y)}{\sum_{y' \in \mathcal{Y}} p_{XY}(x,y')}\,. \tag{43}$$

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For a **joint density**, at values with $f_X(x) > 0$ (and $f_Y(y) > 0$ for the likelihood form),

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- $P(A|B)$, the **posterior** probability, is the probability of A given B, i.e., after B is observed.
- $P(B|A)$ is the probability of observing B given A. As a function of A with B fixed, this is the **likelihood**.

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Two random variables X and Y are **independent** if $F_{XY}(x, y) = F_X(x)F_Y(y)$ for all values of x and y.

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Lemma 61

If X and Y are independent, then for any value sets $\mathcal{A}, \mathcal{B} \subseteq \mathbb{R}$ under consideration,

$$P(X \in \mathcal{A}, Y \in \mathcal{B}) = P(X \in \mathcal{A})P(Y \in \mathcal{B}) \tag{45}$$

Independence

Two random variables X and Y are **independent** if $F_{XY}(x, y) = F_X(x)F_Y(y)$ for all values of x and y.

- For discrete random variables, $p_{XY}(x, y) = p_X(x)p_Y(y)$ for all $x \in \text{Val}(X), y \in \text{Val}(Y)$.
- For discrete random variables, $p_{Y|X}(y|x) = p_Y(y)$ whenever $p_X(x) > 0$, for all $y \in \text{Val}(Y)$.
- For continuous random variables, $f_{XY}(x, y) = f_X(x)f_Y(y)$ (up to equality almost everywhere).
- For continuous random variables, $f_{Y|X}(y|x) = f_Y(y)$ whenever $f_X(x) > 0$, for all $y \in \mathbb{R}$.

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Remark 62

If X is independent of Y, then any function of X is independent of any function of Y.

Expectation and covariance

Definition 63 (Expectation)

Discrete case: Suppose that X, Y are discrete and $g : \mathbb{R}^2 \to \mathbb{R}$ is a function for which the expectation exists. The expected value of $g(X, Y)$ is defined as:

$$\mathbb{E}\left[g(X, Y)\right] \triangleq \sum_{x \in \mathsf{Val}(X)} \sum_{y \in \mathsf{Val}(Y)} g(x, y) p_{XY}(x, y)\,. \tag{46}$$

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Continuous case: For continuous random variables X, Y, the analogous expression is

$$\mathbb{E}\left[g(X,Y)\right] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) f_{XY}(x,y)\, dy\, dx\,. \tag{47}$$

Covariance

Definition 64 (Covariance)

If X and Y have finite second moments, their covariance is

$$\mathrm{Cov}(X, Y) \triangleq \mathbb{E}\left[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\right] = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y] \,. \tag{48}$$

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Diagnostic: Does zero covariance imply independence?

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- If X and Y are independent, then $\mathbb{E}[f(X)g(Y)] = \mathbb{E}[f(X)]\,\mathbb{E}[g(Y)]$ whenever these expectations exist.

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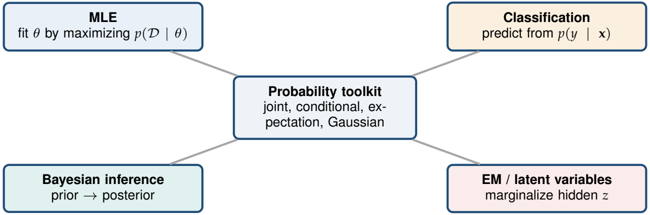
- If X and Y are independent, then $\mathbb{E}\left[f(X)g(Y)\right] = \mathbb{E}\left[f(X)\right]\mathbb{E}\left[g(Y)\right]$ whenever these expectations exist.
- Counterexample: If $X \sim \mathrm{Uniform}(-1,1)$ and $Y = X^2$, they are dependent but uncorrelated.

Where probability returns in this course

Probability toolkit
joint, conditional, ex-
pectation, Gaussian

Goodfellow–Bengio–Courville, 2016, §§3, 5.5, 19.2.

Where probability returns in this course



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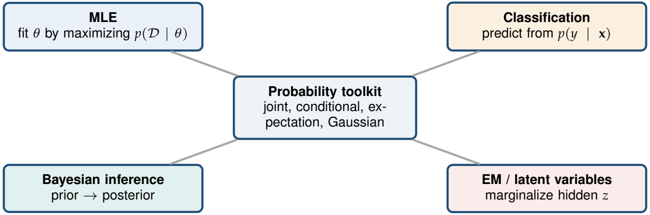
Where probability returns in this course



Carry-forward checklist

Mass, density, or probability? Positive conditioning denominator?
Which variable is marginalized? Which support/moment assumptions apply?

Goodfellow–Bengio–Courville, 2016, §§3, 5.5, 19.2.